



SBI Sentinel

STATE BANK OF INDIA · GFF 2026

● AGENTIC AI & EMERGING TECH · DIGITAL ENGAGEMENT

The Autonomous Financial Wellbeing Engine

An always-on team of AI agents that watches every customer's financial life, catches trouble **before** it happens, and hands them a ready-to-approve plan.

The agents propose. The customer approves. The bank executes.

Human-in-the-loop

Glass-box scoring

RBI + DPDP ready

Deployable inside SBI

Banks **react**. By the time a customer asks for help, the damage is already done.

Today the customer must notice the problem, know the product, and initiate the ask. Most never do, until an EMI bounces, a CIBIL score drops, a family is left under-insured, or a scam clears.



Silent delinquency

An EMI bounce & CIBIL hit are visible only after they happen, never predicted.



Protection gaps

Millions are under-insured & under-invested; no one flags the gap in time.



Fraud after the fact

Scam patterns are caught in reports, not stopped in the moment.



Generic chatbots

Q&A bots answer questions, they don't watch, predict, or act.

50 cr+

SBI customers to serve, reactively today

₹40L cr

Loan book exposed to preventable slippage

Reactive

The status quo: help only when asked

● THE INSIGHT

Others build an AI that talks about your money.
Sentinel builds an AI team that **watches, predicts, plans, proves it's compliant, explains itself, and waits for your yes.**

From product-selling to wellbeing-guarding

Sentinel optimizes the **customer's financial health**, not the bank's product quota. Trust compounds into deposits, loyalty, and engagement.

From autonomous to accountable

A regulated bank can't let AI move money alone. Sentinel is **human-in-the-loop by design**, the single move that makes it deployable inside SBI.

SBI Sentinel, an agentic wellbeing layer inside YONO

Eight specialist AI agents, orchestrated by a supervisor, run a continuous closed loop on each customer, grounded in SBI product knowledge and RBI/DPDP rules.



Detect & Predict

Continuously scores wellbeing and forecasts risk, an EMI bounce, a liquidity crunch, a fraud pattern, days ahead.



Plan & Simulate

Generates 2-3 personalised options and simulates each plan's future impact on the customer's score and cashflow.



Prove & Explain

A deterministic engine certifies every plan against RBI/DPDP; an explainability ledger shows the evidence.

✓ Non-negotiable

The AI never moves money on its own. Every action is consent-gated, reversible, executed on existing SBI rails, and written to an immutable audit ledger. **Agent proposes, customer approves, bank executes.**

● HOW IT WORKS

One closed loop, from a signal to a safe, approved action



Built on **LangGraph** (agent orchestration) + **MCP** (governed tool access) + a **deterministic compliance engine** that can veto any plan.

Eight specialists, one supervisor



Orchestrator

Routes events, owns the plan lifecycle, enforces the human-approval gate.



Financial Health

Computes the glass-box Wellbeing Score & explains every pillar move.



Risk & Early-Warning

Predicts EMI bounce, overdraft, liquidity crunch, NPA slide.



Fraud & Anomaly

Detects scam patterns & anomalies; freezes on consent.



Planner

Generates 2-3 personalised intervention options, never just one.



Simulation

Monte-Carlo projects each plan's future score & cashflow.



Compliance & Guardrail

Deterministic RBI/DPDP + suitability checks. Can veto any plan.



Explainability

Turns the reasoning trail into plain language + an audit record.

Insurance-gap, investment-gap, debt, tax & life-event analysis are **governed tools** the agents call, keeping the architecture clean and auditable.

● GLASS-BOX SCORING

The Financial Wellbeing Score

A single number from **0-1000**, built from six pillars, where **every point is traceable to a transaction**. No black box. This is what satisfies RBI's expectation of explainable AI.

- 0-400 Critical
- 401-600 At-Risk
- 601-800 Stable
- 801-1000 Thriving

612

Rajesh, At-Risk band

→

690

After the approved plan



Any LLM can suggest. Sentinel proves it's safe and shows its work.

① Compliance-Validated Plan

A **deterministic rule engine** (not the LLM) checks every plan against RBI conduct rules, DPDP consent, suitability & mis-selling. It issues a **Compliance Certificate**, or a **VETO**. Unsafe advice never reaches the customer.

✓ Suitability OK

✓ No mis-sell

✓ Consent valid

② Explainability Ledger

Every recommendation carries its **confidence, the exact evidence transactions, the reasoning trail, and the certificate**, a plain-language story for the customer and a tamper-evident record for the auditor.

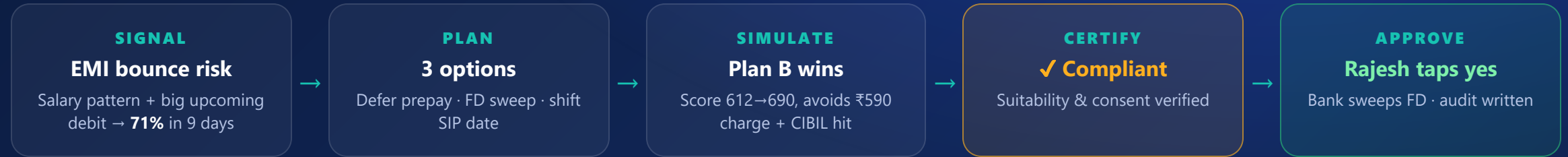
Confidence 0.86

3 evidence txns

Reversible

Why it's defensible: replicating this needs banking-grade regulatory grounding + a governed tool layer + an audit spine, not just a clever prompt. That's a moat a weekend fintech can't cross.

Meet Rajesh, 34, Pune, salaried, home loan, two kids



What Rajesh sees

"Your EMI on the 5th may bounce. Move ₹12,000 from your FD for 6 days and you avoid a ₹590 charge and a credit-score dip. Approve?", with the two transactions that triggered it.

What SBI gets

A prevented delinquency, a delighted customer, a fully audited action, and a fraud alert the same engine raised on a suspicious ₹49,999 transfer, frozen on consent.

Enterprise, bank-grade, data-localized

Channels

YONO · Sentinel UI · RM console to the API Gateway
(OAuth2/OIDC, consent-enforced)

Agent orchestration

LangGraph supervisor + 8 agents · MCP tool servers ·
LLM gateway (frontier-class)

Domain services

Financial engine · Compliance · Consent · Audit ·
Ingestion (FastAPI)

Event & workflow

Kafka event bus · Temporal durable human-in-the-
loop workflows

Data & retrieval

PostgreSQL · Neo4j graph · pgvector/Qdrant · Redis ·
hybrid RAG (RBI/DPDP/SBI corpus)

Platform

Kubernetes on sovereign-cloud/on-prem VPC · India
localization

OpenTelemetry · Prometheus · Grafana

Langfuse LLM tracing & evals

mTLS · RBAC · field-level encryption

PII tokenization · no-log / no-train

Append-only audit

Full C4 diagrams, ER model, agent specs & OpenAPI in the linked GitHub repo ([/docs](#)).

A production stack, not a demo stack

Frontend

React 18 · TypeScript · Vite · Tailwind · shadcn/ui
· Recharts · Framer Motion

Backend

Python 3.12 · FastAPI microservices · Kafka ·
Temporal · Redis

Agentic AI

LangGraph · MCP tool servers · a frontier-class
LLM via provider-abstracted gateway

Guardrails

Deterministic compliance engine (OPA/Rego +
Python) · confidence gating · HITL

Data & retrieval

PostgreSQL · Neo4j · pgvector/Qdrant · hybrid
search (BM25 + dense + rerank)

Platform & Ops

Docker · Kubernetes · sovereign VPC · OTel ·
Prometheus · Grafana · Langfuse

Indic ASR (Bhashini), multilingual

OCR document ingestion

India data localization

DPDP consent ledger

Every agent moves an SBI metric, ethically

CAPABILITY	SBI OUTCOME	WHY IT MATTERS
Risk & Early-Warning	Lower NPA · lower collection cost	Pre-delinquency nudges on a ₹40L-cr book, bps improvements are huge
Fraud & Anomaly	Fraud loss avoided	Real-time detection + consent-based freeze; better RBI reporting SLA
Health + Planner	CASA retention · engagement	A guardian that saves money drives deposits & YONO daily-active use
Gap analysis (consented)	Ethical cross-sell	Suitability-checked SBI Life / SBI MF, conversion up, mis-sell risk zero
Explainability + Compliance	Deployability · trust	Audit-ready, regulator-friendly, the reason this ships, not just demos

↓ **NPA**

Early-warning lead time (days)

↑ **CASA**

Retention & deposits

↓ **Cost**

To serve, via self-service

↑ **DAU**

YONO stickiness

From prototype to 50 crore customers

Now · 0-3 months

Prototype the golden-path loop: FWS engine, Risk + Planner + Compliance + Explainability agents, HITL approve/execute, one YONO-embedded journey. Pilot on a synthetic cohort.

Next · 3-12 months

All 8 agents, fraud + insurance/investment gaps, Neo4j customer-360, Langfuse eval harness, RBI-sandbox pilot with a real branch cohort, DPDP consent ledger live.

Later · 12-24 months

Full YONO rollout, multilingual voice (Bhashini), household-level planning, tax optimization, and B2B2C licensing to cooperative banks & NBFCs.

The asset SBI builds: a compliance-validated, explainable agentic layer, reusable across lending, insurance, wealth, and collections. Sentinel is a platform, not a feature.

Built to score on all six SBI evaluation criteria

CRITERION	HOW SBI SENTINEL ANSWERS IT
Innovation	A compliance-validated plan and an explainability ledger, an agentic layer that proves each action is safe, not just another chatbot.
Technical feasibility	A running React prototype today, with a documented services, data, and agent architecture ready for the 30-day build phase.
Business potential	Tied to real P&L: lower NPA, fraud loss avoided, CASA retention, and suitability-checked cross-sell income.
Scalability	Event-driven microservices on a sovereign VPC, tiered processing designed for 50 crore customers.
User experience	One approval tap, plain-language reasons with evidence, multilingual-ready, embedded in YONO.
Regulatory readiness	Human-in-the-loop, deterministic RBI and DPDP checks that can veto, consent ledger, immutable audit. Built for the risk review.

One idea, three pillars: Sentinel **engages** proactively on behaviour and life events, drives **digital adoption** of SBI payments, investments and insurance through every plan, and opens **acquisition** at the household level, all consent-first.



SBI Sentinel

AUTONOMOUS FINANCIAL WELLBEING ENGINE

SBI stops being a place you transact. It becomes the guardian that watches your back.

Predictive, not reactive. Explainable, not black-box. Human-approved, not autonomous.
Built to deploy inside SBI, and to move NPA, fraud, CASA, and engagement from day one.

The agents propose

The customer approves

The bank executes

Team: Tejas Mane, NFSU Gandhinagar · **Repo:** github.com/tejcodes-rex/sbi-sentinel · SBI Hackathon @ GFF 2026